unext

PHILIPS MR TEAM · AGENTIC AI WORKSHOP · 4–5 NOVEMBER 2026

Participant Prerequisites

Please complete everything on this sheet by **Monday 2 November 2026**. Day 1 opens at 09:00 with a 30-minute environment check — every minute saved there goes into the build. The workshop is fully hands-on on your own laptop: VS Code with GitHub Copilot or OpenAI Codex for spec-driven development, AWS Bedrock for models and Bedrock Guardrails for safety.

Day	What you will do
Day 1 · Wed 4 Nov	Environment check and spec-driven development (Spec → Plan → Tasks → Code). Agent orchestration, MCP, then build one agentic use case: FastAPI on a 10-point scaffold, LangGraph / LangChain / CrewAI agents, an MCP agent, Skills, a database and a Streamlit UI.
Day 2 · Thu 5 Nov	Why guardrails and eval harnesses matter; OWASP LLM Top 10 and OWASP Agentic Top 10. Add Bedrock Guardrails, app-level guard layers and an eval harness, then red-team the Day 1 app with attack scripts and harden it.

1. Laptop and software

Laptop: Windows 10/11, macOS 13+ or Ubuntu 22.04+; 16 GB RAM recommended (8 GB minimum); 10 GB free disk; **admin rights to install software** (or IT pre-install of the items below).

Tool	Version / notes	Verify with
Visual Studio Code	Latest stable	`code --version`
VS Code extensions	Python, Pylance, GitHub Copilot + Copilot Chat, and/or OpenAI Codex	Extensions panel
Python	3.11 or 3.12 (64-bit)	`python --version`
uv (recommended) or pip + venv	Fast, reproducible installs from the starter repo	`uv --version`
Git	2.40+; GitHub account able to clone the workshop repo	`git --version`
AWS CLI v2	Used for SSO login and Bedrock smoke tests	`aws --version`
Node.js	20 LTS or later — for MCP Inspector and promptfoo	`node --version`
Browser	Chrome or Edge — for the Streamlit UI	—
Docker Desktop (optional)	Only if your pod chooses Postgres instead of SQLite	`docker --version`

2. Access and accounts (provided by Philips)

- ☐ **AWS:** SSO / IAM role in the workshop account, in the Philips-approved region, with Bedrock model access enabled and permission for `InvokeModel`, `Converse`, `CreateGuardrail` and `ApplyGuardrail`.
- ☐ **Coding agent:** active GitHub Copilot licence with agent mode allowed by policy, and/or OpenAI Codex — connected to the **Philips-provided OpenAI / Claude API endpoint**.
- ☐ **Secrets:** endpoint URLs and keys go in a local `.env` file only — never in code, chat or Git.
- ☐ **Network:** office network or VPN must reach pypi.org, files.pythonhosted.org, github.com, registry.npmjs.org, AWS endpoints (sts, sso, bedrock, bedrock-runtime) and your Copilot / Codex endpoints. Please test from the network you will use on the day.

Trainer access: Philips MR team to provide the trainer with AWS, Bedrock model and Guardrails access, plus a demo endpoint, by **Tuesday 3 November 2026** so every lab can be run end to end in your environment beforehand.

3. Self-check — run before 2 November

The starter-repo link will be emailed to you. Then, in a VS Code terminal:

Command	Purpose
`git clone <starter-repo-url> && cd agentic-ai-workshop`	Get the code
`uv sync`	Installs pinned Python packages (≈ 5 min)
`aws sso login --profile philips-workshop`	Or the credential method Philips specifies
`aws sts get-caller-identity --profile philips-workshop`	Shows the workshop role
`uv run python scripts/env_check.py`	Expect 6 × PASS

env_check.py confirms: toolchain, coding agent, AWS identity, one Bedrock Converse call, one ApplyGuardrail call, and that the empty scaffold's tests pass. If anything fails, reply to the invitation email with the output.

4. Local configuration (.env)

Copy `.env.example` to `.env` and fill in the values Philips shares. `.env` is already in .gitignore.

Variable	Example	Source
`AWS_PROFILE`	philips-workshop	Philips cloud team
`AWS_REGION`	Philips-approved region	Philips cloud team
`BEDROCK_MODEL_ID`	Model or inference-profile ID	Philips cloud team
`BEDROCK_GUARDRAIL_ID / _VERSION`	Created by you on Day 2	Workshop lab
`OPENAI_* / ANTHROPIC_* endpoint + key`	Only if your coding agent reads them from env	Philips (coding endpoints)

5. Common setup issues and quick fixes

Symptom	Fix
pip / uv times out or SSL error	Set HTTPS_PROXY and the corporate CA bundle (SSL_CERT_FILE / UV_NATIVE_TLS=1); ask IT for the proxy values.
Copilot chat has no Agent mode	Update VS Code and the Copilot extensions; check that your organisation policy enables agent mode.
AccessDeniedException from Bedrock	Model access not enabled for that model / region, or the role lacks bedrock:InvokeModel — send the error to the Philips cloud team.
ExpiredToken / SSO session expired	Run `aws sso login --profile philips-workshop` again.
Windows: venv activation blocked	Use uv run … (no activation needed), or Set-ExecutionPolicy -Scope CurrentUser RemoteSigned.

6. What the starter repo installs

Pinned in the repo — no manual installs needed: fastapi, uvicorn, pydantic v2, langchain, langchain-aws, langgraph, crewai, mcp (FastMCP), boto3, streamlit, sqlalchemy, pytest, ruff, mypy; red-team tools garak and PyRIT (Python) and promptfoo (via npx).

7. Skills expected and optional pre-reading

- Comfortable reading and editing Python (functions, classes, type hints; async basics help), using Git and a terminal, and working with REST / JSON.
- **Optional, 30 minutes:** OWASP Top 10 for LLM Applications (2025) · OWASP Top 10 for Agentic Applications (2026) · Model Context Protocol introduction (modelcontextprotocol.io) · Amazon Bedrock Guardrails overview (AWS docs).

8. Ground rules for the two days

- **Synthetic data only** — no patient data, PHI, customer data or production systems in any lab.
- Red-team only your assigned pod's local build and the workshop AWS account — never Philips production endpoints.
- Work in pods of three with rotating roles (spec owner · builder · reviewer); bring your charger.

READY FOR 4 NOVEMBER? Tick all six before Monday 2 November.

- ☐ Software in section 1 installed
- ☐ Copilot / Codex answers via Philips endpoint
- ☐ env_check.py shows 6 × PASS
- ☐ AWS SSO login works
- ☐ Starter repo cloned and uv sync done
- ☐ Charger packed